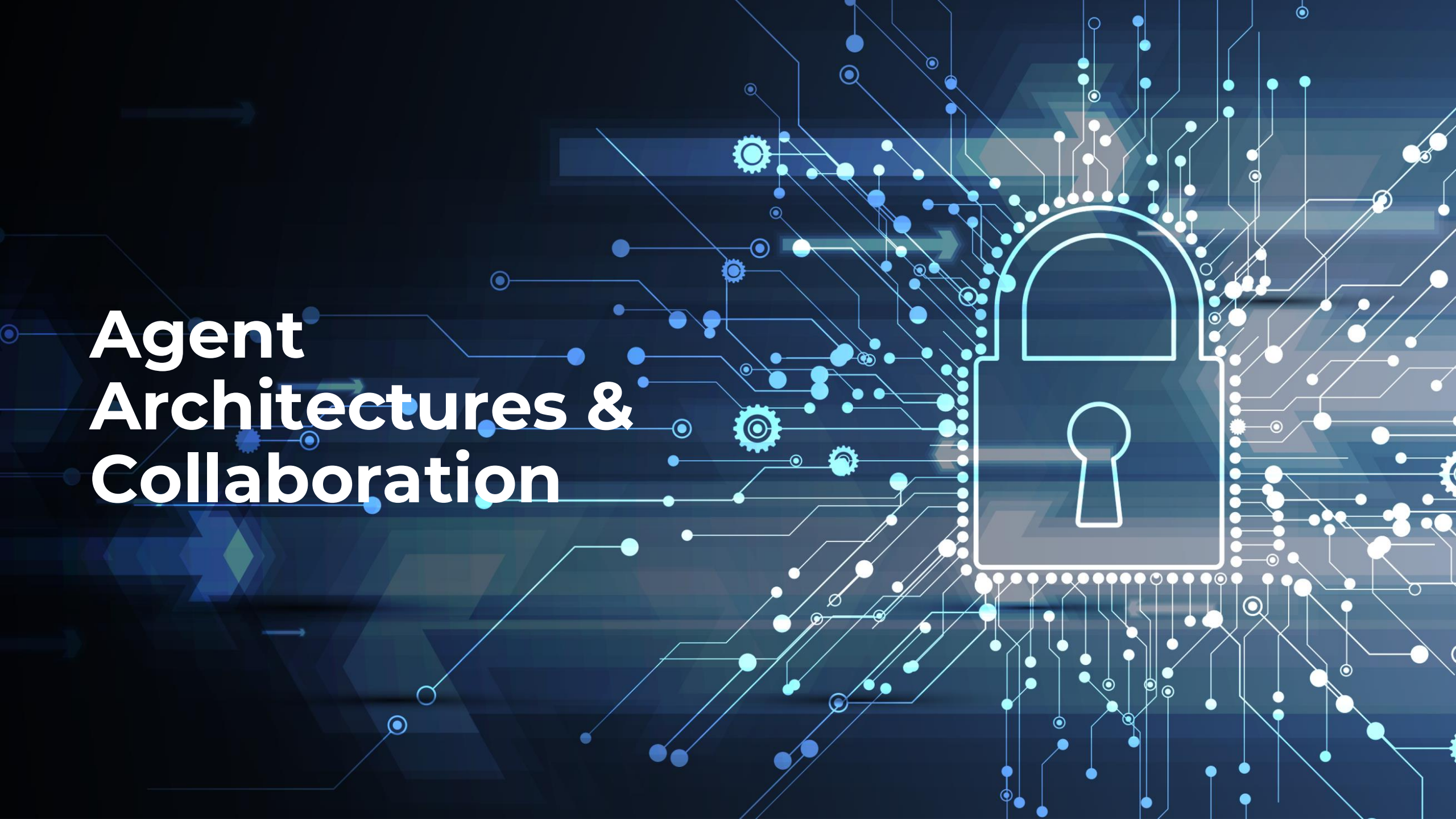
IITM PRAVARTAK
CATALYSING INNOVATION

Professional Certificate Programme in Agentic AI and Applications

Agent Architectures & Collaboration



Revisit the Learning So Far

Let’s take a quick look at the key ideas and concepts you have explored in your learning journey.

- Understanding the AI ecosystem, including agents, functions and APIs
- How the agent life cycle progresses from initial goals to continuous growth
- An overview of agentic frameworks such as LangChain, Autogen and CrewAI
- Core principles that guide tool integration within AI agent frameworks
- How AI agents can be extended and enhanced using external APIs

Topics Covered

- Agentic AI System Architectures
- Types of AI Agents: Fundamentals & Industry Applications
- Task–Oriented v. Goal-Oriented Agents & Agent Communication Frameworks

Agentic AI System Architectures

Agentic AI System Architectures

- Range from reactive responses to multi-agent orchestration
- Support specific use cases and complexity levels in modern AI systems

The Architecture Spectrum

Agentic AI systems span a spectrum of complexity and autonomy

Architectural choices depend on task complexity, autonomy level and multi-agent coordination needs

The Architecture Spectrum

Simple & fast

Reactive patterns for immediate responses

Workflow focus

Deliberative approaches for complex decisions

Distributed intelligence

Multi-agent systems for coordinated tasks agents

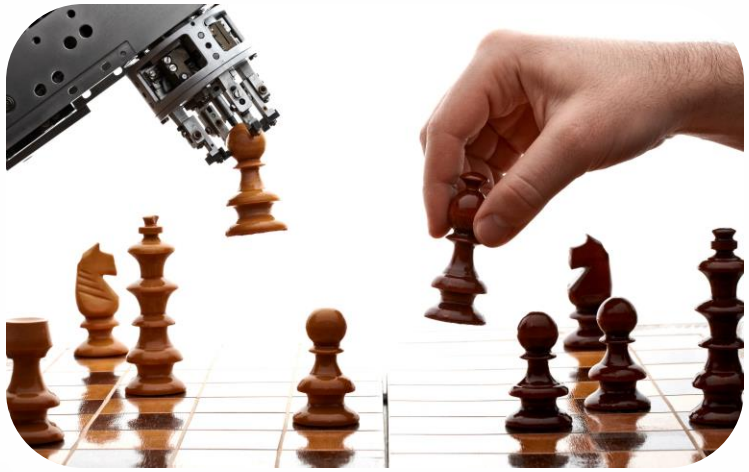
Reactive Agents: The Thermostat Approach

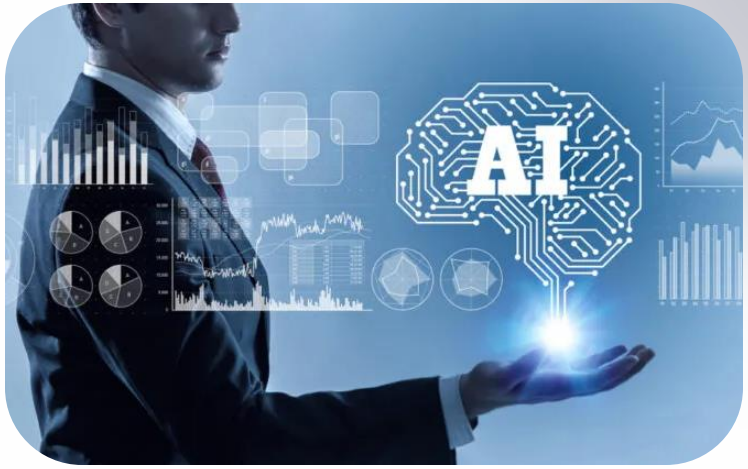
Stimulus-Response Pattern

Best for: Real-time systems requiring instant responses with predictable patterns

- Operate on a simple principle: "If X happens → do Y."
- Do not store history or plan ahead, enabling fast, efficient reactions
- Ideal for real-time tasks, such as chatbots, trading bots and robotic sensors
- Excel at predictable responses, like thermostats adjusting to temperature changes

Deliberative Agents: The Chess Master





- Maintain internal models and plan actions using symbolic reasoning
- Excel in strategic scenarios, such as autonomous navigation and complex problem-solving
- Excel in tasks requiring foresight and strategic thinking
- Used in autonomous vehicles for route optimisation in expert systems for complex domains like medical diagnosis and financial planning

Trade-off: Deliberative agents sacrifice speed for sophistication, making slower but more informed decisions by considering context and long-term impact.

Hybrid Agents: Best of Both Worlds

Reactive layer

Instant responses to critical situations like:

- Emergency braking
- Collision avoidance
- Real-time adjustments

Deliberative layer

Strategic planning and reasoning like:

- Route optimisation
- Resource allocation
- Long-term goals

Hybrid Agents: Best of Both Worlds

- Self-driving cars combine deliberative planning for optimal routing with reactive responses to dynamic hazards
- This dual-layer design enables strategic foresight alongside rapid, instinctive action

BDI Agents: Human-Like Reasoning

Beliefs

Represents the agent's knowledge of the world and environment

Desires

Defines the objectives and goals the agent aims to achieve

Intentions

Tracks plans and actions committed to achieving those goals.

Mirror human reasoning, by updating beliefs, evaluating desires and committing to intentions

Act contextually, such as allocating resources when deadlines approach to ensure project success

Multi-Agent Systems: Team Intelligence

- Represent distributed intelligence, where multiple agents interact via established communication protocols
- Each agent maintains specialised capabilities, while contributing to collective objectives

Multi-Agent Systems: Team Intelligence

Cooperative

Agents collaborate toward shared goals, like drone swarms in search and rescue

Competitive

Agents compete for resources within rules, as seen in algorithmic trading

Coordinated

Agents perform specialised roles with orchestrated actions, like in supply chains

Learning & Tool-Using Agents

Learning agents

Improve performance through feedback loops, using reinforcement learning to refine decisions over time

- Continuously refine behaviour, like a piano student responding to feedback
- Applied in Game AI, adaptive tutoring and optimisation bots

Tool-Using agents

Extend capabilities by calling external APIs and functions, selecting the right tool for each task

- Leverage external resources intelligently, like a handyman choosing tools with precision
- Used in LangChain agents, API-driven assistants and computational tools

Architecture Decision Matrix

Architecture pattern	Speed	Complexity	Ideal use cases
Reactive	⚡ Fast	◆ Simple	Real-time chatbots, trading bots, sensor systems
Deliberative	🐌 Slow	◆ ◆ ◆ Complex	Strategic planning, knowledge systems, route optimisation
Hybrid	⚡🐌 Mixed	◆ ◆ ◆ ◆ Very complex	Autonomous vehicles, adaptive assistants, robotics
Multi-agent	⚡ Fast	◆ ◆ ◆ ◆ ◆ Extremely complex	Supply chains, drone swarms, distributed systems

Key Architectural Insights

Real-world autonomous systems blend reactive speed with deliberative planning for optimal performance in dynamic settings

Modern agents extend capabilities via APIs and external tools, making tool-using patterns essential

Multi-agent orchestration enables horizontal scaling by distributing intelligence across specialised components

Hybrid is king

Tool integration

Scalability matters

Choose architecture by task needs, not trends—the best agent solves your problem effectively.

Types of AI Agents: Fundamentals & Industry Applications

(Topic 2 of 3)

Types of AI Agents

- Four core types of AI agents underpin real-world applications across diverse industries
- Architectures range from simple reactive systems to complex multi-agent orchestrations

Agent Architecture Overview

- Perceive their environment
- Take goal-directed actions

AI agents are autonomous systems that

- System performance
- Scalability
- Complexity

Agent architecture determines

Agent Architecture Overview

Environmental perception: Agents collect data from sensors, APIs, databases or users to assess context.

Action execution: Agents act via API calls, robotic movements, data updates or system communication.

Decision processing: Agents apply reactive rules, planning or hybrid logic to process information.

Learning & adaptation: Advanced agents learn from outcomes to refine decisions and adapt to change.

Reactive Agents: Stimulus-Response Systems

Core Characteristics

Reactive agents operate on direct stimulus-response patterns without internal models or memory

- They execute predetermined actions based on "**if X then Y"** behavioural rules in response to environmental inputs:
  - No internal world model or memory
  - Immediate response to environmental stimuli
  - Simple rule-based decision making
  - Minimal computational overhead
  - Fast execution but limited adaptability

Key advantage: Speed and simplicity make reactive agents ideal for time-critical tasks requiring millisecond responses.

Reactive Agents in Industry

Error handling

- Assembly-line robots stop instantly when sensors detect obstacles
- No planning—just immediate response to sensor data

High-frequency trading

- Trading bots execute buy/sell orders in microseconds based on price movements
- Speed is everything—deliberation risks missed opportunities

Smart home systems

- Thermostats adjust temperature based on sensor readings
- Simple but effective — Heating activates immediately when temperature drops below threshold

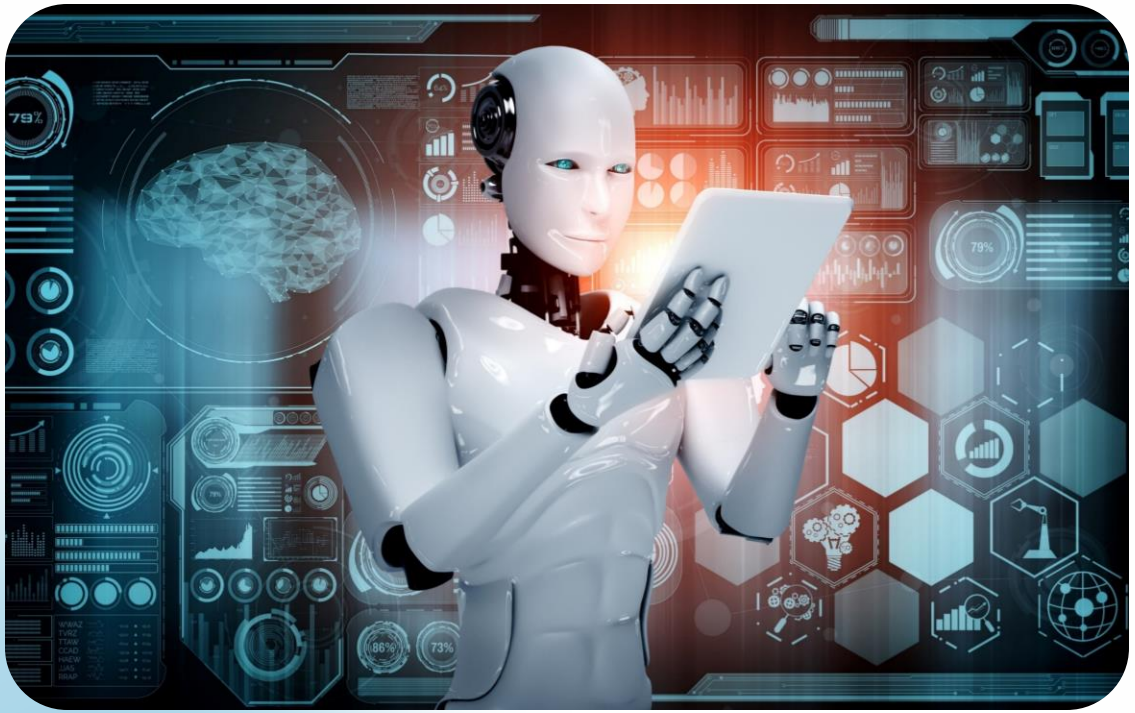
Reactive Agents in Industry

Training Insight

Prioritise reactive agents when rapid response outweighs strategic reasoning

Apply in safety systems and real-time control tasks

Deliberative Agents: Model-Based Planning



- Maintain comprehensive internal models and plan strategically before executing actions
- Simulate potential outcomes and select optimal paths to achieve goals

Deliberative Agents: Model-Based Planning

World model maintenance

Continuously update internal models with obstacles, resources and dynamic conditions

Strategic planning

Simulate "what-if" scenarios, analyse action sequences and optimise for long-term goals

Deliberate execution

Execute with confidence, monitor outcomes and adjust plans from real-world feedback

Deliberative Agents: Industry Applications

Logistics optimisation

Route-planning AI analyses traffic, delivery windows, and vehicle capacity to optimise schedules before dispatch.

Healthcare diagnosis

Medical AI analyses history, symptoms, and test results to recommend treatments aligned with guidelines and drug interactions.

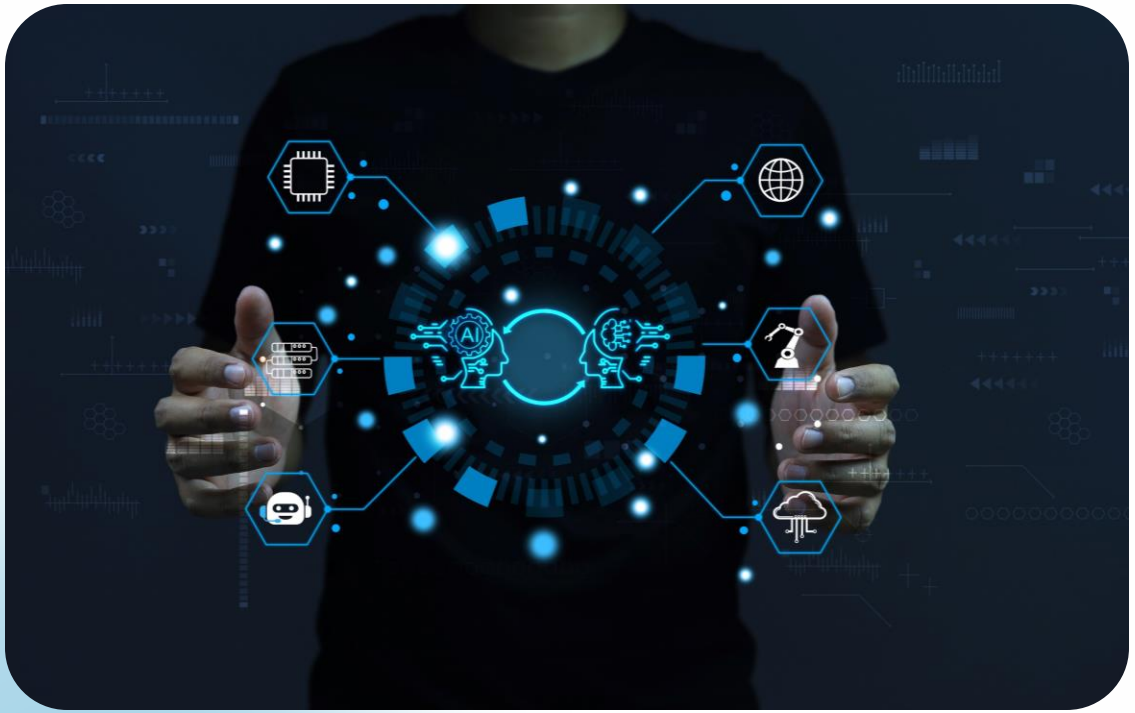
Deliberative Agents: Industry Applications

Key Strengths

Deliver superior accuracy and foresight through comprehensive analysis

Excel when strategic decision-making outweighs response speed requirements

Hybrid Agents: Best of Both Worlds



- Combine strategic planning with real-time responsiveness of reactive systems
- Address limitations of pure models to perform reliably in dynamic environments

Hybrid Agents: Best of Both Worlds

Deliberative layer computes optimal path using traffic, weather and destination data.

Updates planning models and reactive responses using experience and feedback

Reactive layer manages obstacles, lane changes and emergency braking

Route planning

Continuous learning

Real-time adaptation

Hybrid Agents: Best of Both Worlds

Autonomous vehicles exemplify hybrid architecture by planning routes strategically while reacting instantly to pedestrians, signals, and road conditions.

Multi-Agent Systems: Distributed Intelligence

System Architecture

Multi-agent systems deploy specialised agents that coordinate distinct tasks to solve complex problems and achieve shared goals.

Distributed processing eliminates single points of failure

Specialised agents handle domain-specific tasks

Communication protocols enable coordination

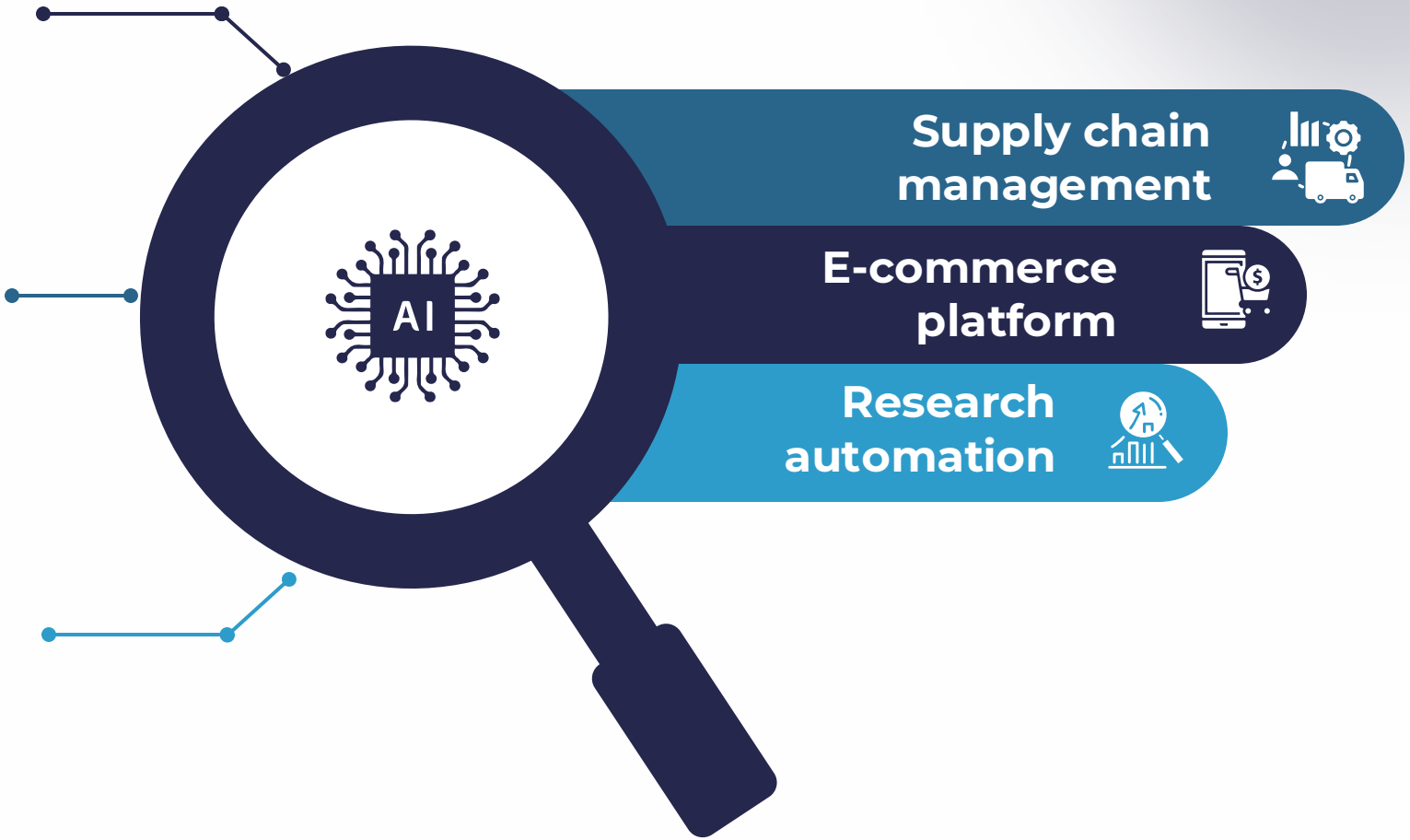
Scalable architecture grows with problem complexity

Multi-Agent Systems: Distributed Intelligence

Agents coordinate inventory management, shipping optimisation and demand forecasting.

Recommendation agent suggests products, inventory agent manages stock, payment agent handles transactions.

Agents read papers, extract insights and draft reports in a coordinated workflow.



Choosing the Right Agent Architecture

<1ms
Reactive response time

Ideal for safety-critical systems requiring instant responses

95%
Deliberative accuracy

Superior decision quality through comprehensive analysis

3x
Hybrid performance

Industry adoption rate compared to pure architectures

Choosing the Right Agent Architecture

Architecture Selection Guide

Reactive

Time-critical systems (trading bots, safety controls)

Deliberative

Strategic applications (route planning, medical diagnosis)

Hybrid

Complex real-world systems (autonomous vehicles, advanced robotics)

Multi-agent

Large-scale distributed problems (supply chains, research teams)

Choosing the Right Agent Architecture

Key Points to Remember

- Reactive agents are firefighters — Fast reaction
- Deliberative agents are chess players — Strategic planning
- Hybrid agents are race car drivers — Plan the race, react to conditions
- Multi-agent systems are orchestras — Coordination plus specialisation

Task-Oriented v. Goal-Oriented Agents & Agent Communication Frameworks

Exploring the evolution of intelligent agents from simple task executors to sophisticated goal-driven systems that communicate, reason and adapt autonomously.

The Evolution of Artificial Intelligence

Rule-based systems

Early AI systems followed predetermined rules and decision trees to solve specific problems

Goal-oriented intelligence

Modern agents that pursue broader objectives through dynamic planning and adaptive reasoning

Task-oriented agents

Agents complete specific, well-defined tasks through structured workflows.

Collaborative networks

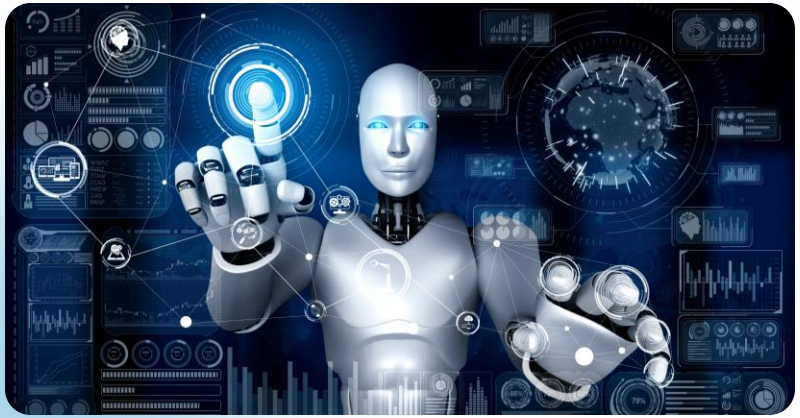
Multi-agent systems communicate and coordinate to achieve complex shared goals.

Defining the Agents

Task-Oriented v. Goal-Oriented

Task-oriented agents

- Complete predefined tasks step-by-step using reactive, rule-based approaches
- Excel at structured workflows but lack flexibility



Goal-oriented agents

- Pursue broader objectives through dynamic planning and adaptive reasoning
- Reason about future states and optimise strategies autonomously

- **Real-world example:** A task-oriented chatbot follows scripts to answer FAQs, while a goal-oriented agent autonomously negotiates and plans workflows.

Task-Oriented Agents in Action

Input recognition

Agent categorises user input using predefined patterns and keywords

Rule application

System applies matching rules or decision trees to select a response

Task execution

Agent executes the structured workflow or delivers a scripted reply

Output delivery

System delivers the output and awaits the next input within task boundaries

Key Features & Differences

Task-Oriented

- **Fixed workflows** — Executes set sequences with minimal deviation
- **Limited reasoning** — Operates within immediate task scope only
- **Reactive behaviour** — Responds via predefined triggers and patterns

Goal-Oriented

- **Dynamic planning** — Adapts strategies by reasoning about future states
- **Memory & learning** — Improves over time using memory and learning systems
- **Proactive pursuit** — Balances objectives, resources, and constraints autonomously

Goal-Oriented Intelligence Framework

Goal setting: Define objectives and success criteria

Planning: Create strategies and action sequences

Execution: Implement plans with real-time monitoring

Assessment: Evaluate progress and outcomes

Adaptation: Adjust strategies based on results

Agent Communication

From Message Passing to Goal-Driven Dialogue

Traditional communication

Focuses on message fidelity, bandwidth and reliable data transmission between systems

Goal-oriented communication

Ties messages directly to task success and shared objectives, optimising relevance under resource constraints

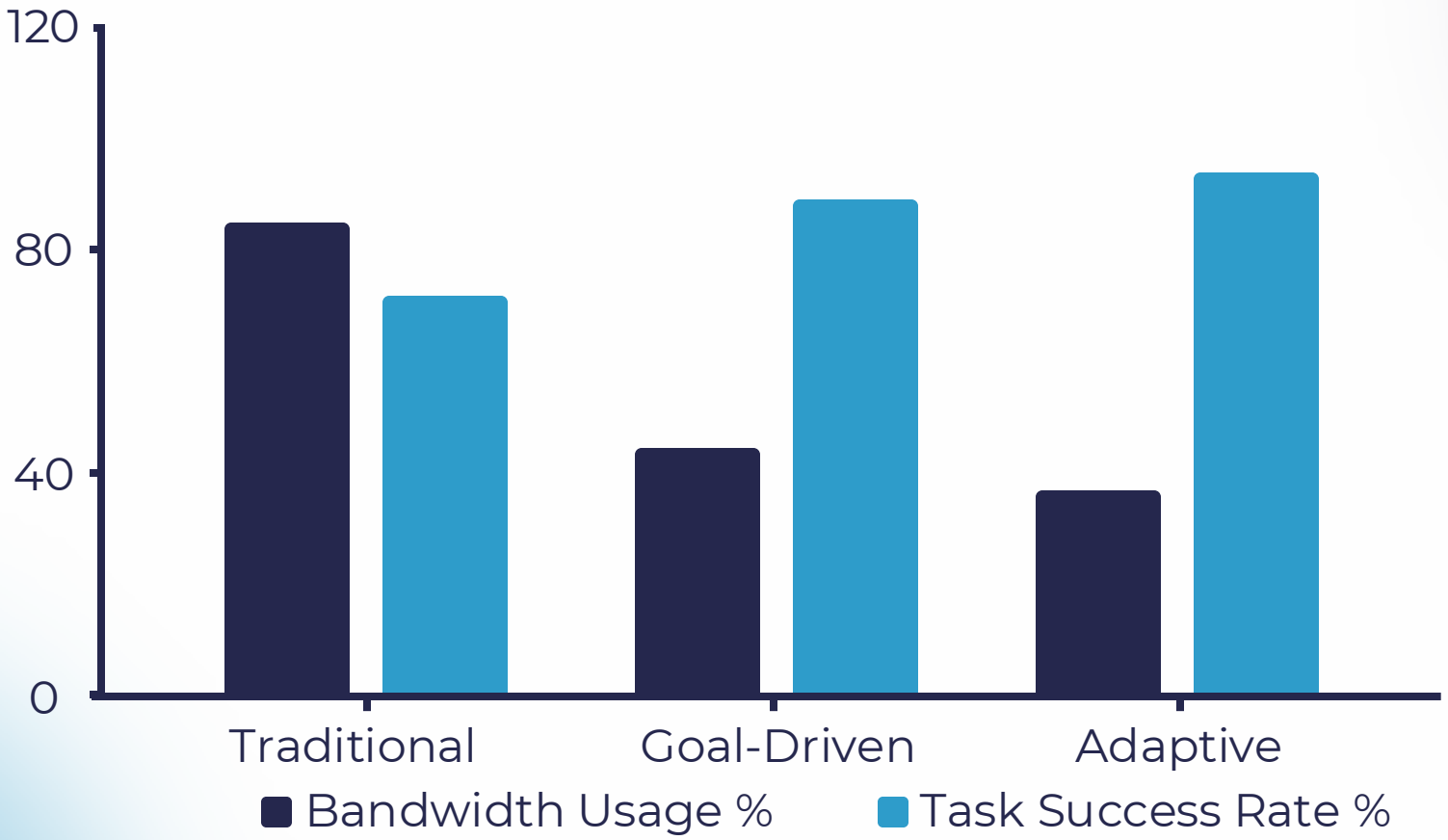
"Goal-oriented communication optimises information exchange based on task relevance rather than just message completeness."

Agent Communication

Example

Swarm robots exchange only goal-critical data, minimising communication while preserving mission effectiveness.

Communication Efficiency Comparison



Goal-driven communication boosts task success while minimising bandwidth, showcasing context-aware efficiency.

Reasoning Frameworks Empowering Agents

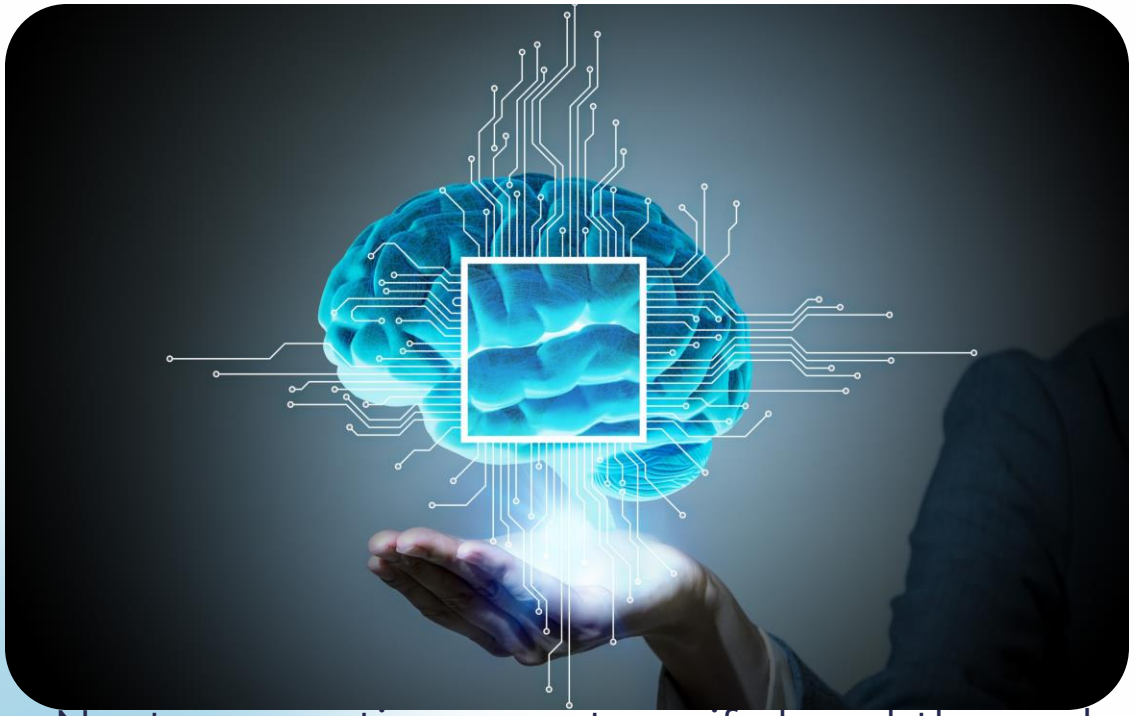
Planning algorithms: Use probabilistic models and reinforcement learning to select optimal actions under uncertainty

Memory integration: Integrate tools and memory systems to support complex decision-making and retain knowledge

Framework evolution: Employ LangChain and LlamaIndex to build autonomous, goal-driven workflows with enhanced capabilities

Conversational AI: Evolve from reactive bots to proactive, goal-driven dialog systems using advanced models like SocialAGI's GDAD

Breakthrough Technologies Enabling Smart Agents



Next-generation agents unify breakthrough technologies to achieve unmatched autonomy and intelligence.

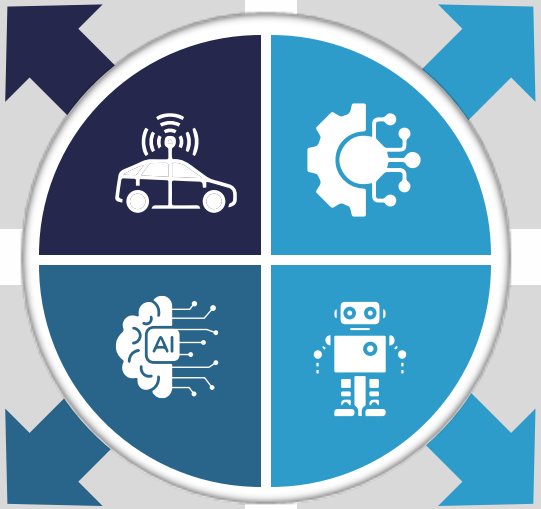
- **Advanced neural networks**: Enable sophisticated reasoning and planning via transformers and attention mechanisms
- **Persistent memory systems**: Maintain context and support learning through long-term and working memory
- **Tool integration frameworks**: Extend agent capabilities beyond language via APIs and external tools

The Future

Dynamic, Collaborative and Context-Aware Agents

Autonomous vehicles
Self-driving systems that navigate and coordinate in real-time

Edge intelligence
Distributed processing at network edges for rapid response



Advanced conversational AI
Context-aware dialogue systems with deep understanding

Multi-agent robotics
Coordinated robot swarms for complex missions

The shift from task execution to goal achievement enables smarter, adaptive agents for dynamic, resource-limited environments.

Q & A

Thank you